Appendix C. CALCULUS

C.1. Boundaries. Let $U \subset \mathbb{R}^n$ be open and bounded, $k \in \{1, 2, \dots\}$.

DEFINITION. We say the boundary ∂U is C^k if for each point $x^0 \in \partial U$ there exist $r > 0$ and a C^k function $\gamma : \mathbb{R}^{n-1} \to \mathbb{R}$ such that—upon relabeling and reorienting the coordinates axes if necessary—we have

$$U \cap B(x^0, r) = \{ x \in B(x^0, r) \mid x_n > \gamma(x_1, \dots, x_{n-1}) \}.$$

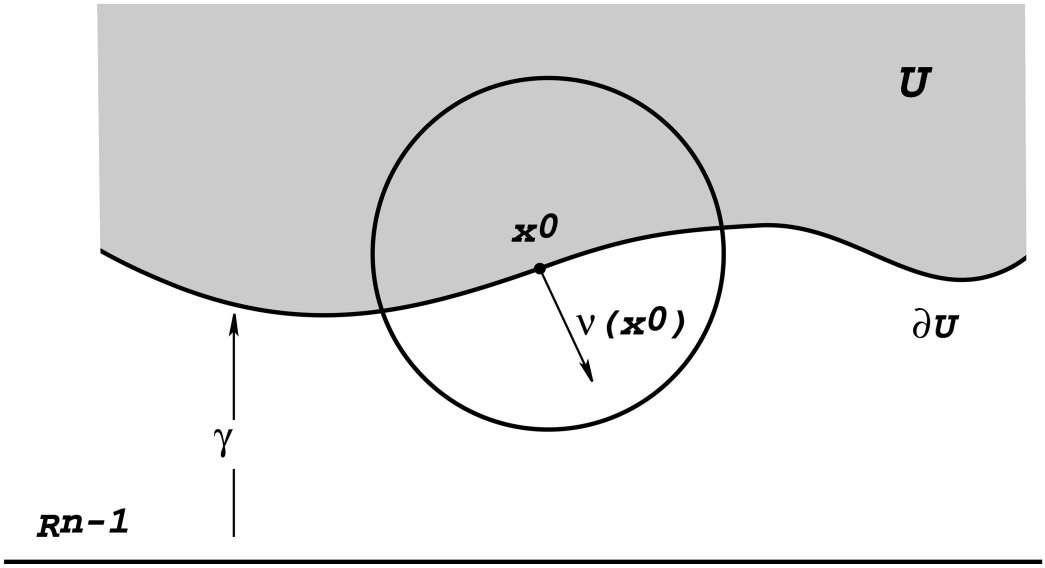
Likewise, ∂U is C^∞ if ∂U is C^k for $k = 1, 2, \dots$, and ∂U is *analytic* if the mapping γ is analytic.

DEFINITIONS.

(i) If ∂U is C^1, then along ∂U is defined the outward pointing *unit normal vector field*

$$\boldsymbol{\nu} = (\nu^1, \dots, \nu^n).$$

The *unit normal* at any point $x^0 \in \partial U$ is $\boldsymbol{\nu}(x^0) = \nu = (\nu_1, \dots, \nu_n)$.



The boundary of U

(ii) Let $u \in C^1(\bar{U})$. We call

$$\frac{\partial u}{\partial \nu} := \boldsymbol{\nu} \cdot Du$$

the (*outward*) *normal derivative* of u.

Straightening the boundary. We will frequently need to change coordinates near a point of ∂U so as to "flatten out" the boundary. To be more specific, fix $x^0 \in \partial U$, and choose r, γ, etc. as above. Define then

$$\begin{cases} y_i = x_i =: \Phi^i(x) & (i = 1, \dots, n-1) \\ y_n = x_n - \gamma(x_1, \dots, x_{n-1}) =: \Phi^n(x), \end{cases}$$

and write

$$y = \boldsymbol{\Phi}(x).$$

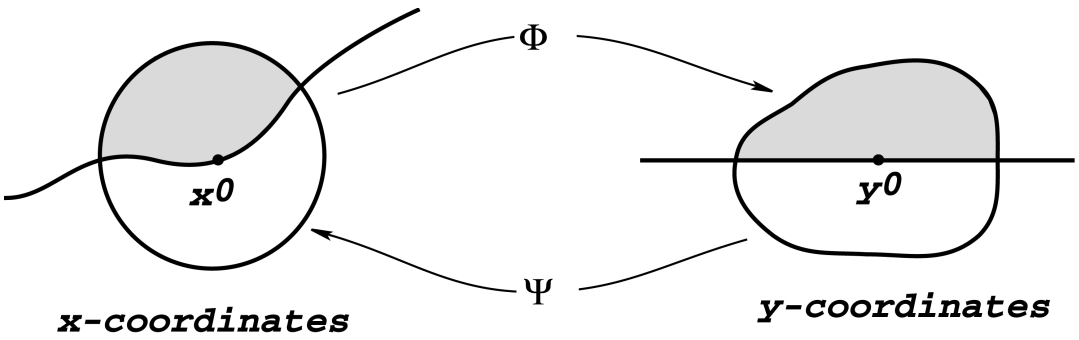
Similarly, we set

$$\begin{cases} x_i = y_i =: \Psi^i(y) & (i = 1, \dots, n-1) \\ x_n = y_n + \gamma(y_1, \dots, y_{n-1}) =: \Psi^n(y), \end{cases}$$

and write

$$x = \boldsymbol{\Psi}(y).$$

Then $\boldsymbol{\Phi} = \boldsymbol{\Psi}^{-1}$, and the mapping $x \mapsto \boldsymbol{\Phi}(x) = y$ "straightens out ∂U" near x^0. Observe also that $\det D\boldsymbol{\Phi} = \det D\boldsymbol{\Psi} = 1$.



Straightening out the boundary

C.2. Gauss–Green Theorem. In this section we assume U is a bounded, open subset of $\mathbb{R}^n$ and ∂U is C^1.

THEOREM 1 (Gauss–Green Theorem)**.**

(i) *Suppose $u \in C^1(\bar{U})$. Then*

$$\int_U u_{x_i}\,dx = \int_{\partial U} u\nu^i\,dS \quad (i = 1, \dots, n). \tag{1}$$

(ii) *We have*

$$\int_U \operatorname{div}\mathbf{u}\,dx = \int_{\partial U} \mathbf{u}\cdot\nu\,dS \tag{2}$$

for each vector field $\mathbf{u} \in C^1(\bar{U};\mathbb{R}^n)$.

Assertion (ii), also called the *Divergence Theorem*, follows from (i) applied to each component of $\mathbf{u} = (u^1, \dots, u^n)$.

THEOREM 2 (Integration by parts formula)**.** *Let $u, v \in C^1(\bar{U})$. Then*

$$\int_U u_{x_i} v\,dx = -\int_U u v_{x_i}\,dx + \int_{\partial U} u v \nu^i\,dS \quad (i = 1, \dots, n). \tag{3}$$

Proof. Apply Theorem 1(i) to uv. □

THEOREM 3 (Green's formulas)**.** *Let $u, v \in C^2(\bar{U})$. Then*

(i) $\int_U \Delta u\,dx = \int_{\partial U} \frac{\partial u}{\partial \nu}\,dS$,

(ii) $\int_U Dv\cdot Du\,dx = -\int_U u\Delta v\,dx + \int_{\partial U} \frac{\partial v}{\partial \nu} u\,dS$,

(iii) $\int_U u\Delta v - v\Delta u\,dx = \int_{\partial U} u\frac{\partial v}{\partial \nu} - v\frac{\partial u}{\partial \nu}\,dS$.

Proof. Using (3), with u_{x_i} in place of u and $v \equiv 1$, we see

$$\int_U u_{x_i x_i}\,dx = \int_{\partial U} u_{x_i}\nu^i\,dS.$$

Sum $i = 1, \dots, n$ to establish (i).

To derive (ii), we employ (3) with v_{x_i} replacing v. Write (ii) with u and v interchanged and then subtract, to obtain (iii). □

C.3. Polar coordinates, coarea formula. Next we convert n-dimensional integrals into integrals over spheres.

THEOREM 4 (Polar coordinates)**.**

(i) *Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuous and summable. Then*

$$\int_{\mathbb{R}^n} f\,dx = \int_0^\infty \left(\int_{\partial B(x_0,r)} f\,dS \right) dr$$

for each point $x_0 \in \mathbb{R}^n$.

(ii) *In particular*

$$\frac{d}{dr}\left(\int_{B(x_0,r)} f\,dx \right) = \int_{\partial B(x_0,r)} f\,dS$$

for each $r > 0$.

Theorem 4 is a special case of the following theorem.

THEOREM 5 (Coarea formula)**.** *Let $u : \mathbb{R}^n \to \mathbb{R}$ be Lipschitz continuous and assume that for a.e. $r \in \mathbb{R}$ the level set*

$$\{ x \in \mathbb{R}^n \mid u(x) = r \}$$

is a smooth, $(n-1)$-dimensional hypersurface in $\mathbb{R}^n$. Suppose also $f : \mathbb{R}^n \to \mathbb{R}$ is continuous and summable. Then

$$\int_{\mathbb{R}^n} f|Du|\,dx = \int_{-\infty}^\infty \left(\int_{\{u=r\}} f\,dS \right) dr.$$

Theorem 4 follows from Theorem 5 if we take $u(x) = |x - x_0|$. Consult [**E-G**, Chapter 3] for more on the coarea formula. The word "coarea" is pronounced, and sometimes spelled, "co-area".

C.4. Moving regions. Consider a family of smooth, bounded regions $U(\tau) \subset \mathbb{R}^n$ that depend smoothly upon the parameter $\tau \in \mathbb{R}$. Write $\mathbf{v}$ for the velocity of the moving boundary $\partial U(\tau)$ and ν for the outward pointing unit normal.

THEOREM 6 (Differentiation formula for moving regions)**.** *If $f = f(x, \tau)$ is a smooth function, then*

$$\frac{d}{d\tau} \int_{U(\tau)} f\,dx = \int_{\partial U(\tau)} f\,\mathbf{v} \cdot \nu\,dS + \int_{U(\tau)} f_\tau\,dx.$$

See Chorin–Marsden [**C-M**] for a proof.